Where Workers Come From: An Occupational Flow Account for the United States, and the Instrument That Makes It Measurable

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Extended summary

Workforce policy in the United States is organised around training. States fund nursing school places, teacher preparation programmes, apprenticeships and community college capacity on the understanding that an occupation short of people is an occupation short of graduates. This paper builds a complete account of how people actually enter and leave every occupation in the country, and finds that arrivals from education are the smallest of the inflows by a wide margin. Across the occupations the data can support, arrivals from another occupation run between four and twelve times arrivals from education, and they occur at every age rather than concentrating among the young, which rules out the possibility that they are graduates recorded under the wrong heading.

The account rests on a measurement result that has to come first, because the obvious way to build it does not work. Occupational mobility is usually measured by following a person across two interviews and comparing the occupation recorded on each. The Current Population Survey assigns occupation codes from verbatim descriptions of what a person does, and two independent codings of the same unchanged job disagree in 48.1 per cent of cases at the detailed level and in 31.8 per cent across only twenty three major groups. That figure is measured here on 114,491 people whose occupation the survey itself records as unchanged and who worked all fifty two weeks of the reference year, with the person link validated against state of residence and sex, which disagree in 0.00 per cent of cases, and against education, which disagrees in 12.6. A linked measure of annual mobility built on that foundation returns 50.0 per cent, which is lower than the disagreement between two codings of the same period, so genuine movement is invisible beneath the noise.

The retrospective design used here avoids the problem for a specific reason. The March supplement asks in a single interview what a person does now and what they did last year, so both codes are assigned at one sitting from one respondent's descriptions, their errors are correlated, and the correlation cancels when the difference between them is taken. The same instrument that looks weaker, because it relies on recall, is the one that works, and I take some care to show why rather than asserting it.

On that basis the paper builds an account of arrivals and departures for 230 occupations,

annually from 2009 to 2025, disaggregated by age band, with a standard error on every figure computed from the 160 replicate weights the Census Bureau publishes for this survey. The account balances exactly rather than approximately, because every person is counted once on each side of the identity from a single source, and nothing in it is imputed or calibrated to an external total.

Three findings follow. Arrivals from another occupation dominate arrivals from education for 98 per cent of occupations, and the dominance survives aggregation to the twenty three major groups at which a coder cannot confuse a nurse with a physical therapist. Those arrivals are flat across the age distribution, so they are career movement rather than misrecorded entry from training. And most of that movement is two way traffic that cancels: only between 15 and 36 per cent of gross lateral flow carries direction, depending on how finely occupations are cut, so the gross movement through an occupation is several times the net change in its size. Shortage is a net phenomenon while the levers policy reaches are gross ones, and the distance between those two quantities is the practical result of the paper.

1 Introduction

An occupation that cannot find enough people is usually treated as an occupation that has not trained enough of them, and the remedy follows from the diagnosis. The difficulty is that the diagnosis has rarely been checked against a full account of where an occupation's people come from, because no such account exists at the level of detail policy operates on. Employment statistics report stocks with great precision and considerable frequency, and they say nothing about the flows that produced them, so an occupation that is stable in headcount and an occupation that replaces a fifth of its people every year are recorded identically.

The flows have been studied, and the difficulty is that the literature which studies them and the literature which studies shortages have not met. Work on gross worker flows established that net employment change is the small residual of much larger movements between employment, unemployment and inactivity (Davis and Haltiwanger, 2014), and work on occupational mobility established that people change occupation often enough for occupation specific human capital to matter for wages (Kambourov and Manovskii, 2008). More recent work has begun to treat the transitions between occupations as a network with its own structure (Cheng and Park, 2020). None of this has been assembled into an account that a workforce planner could open and read off, for a named occupation, how many people arrived from training, how many arrived from another occupation, how many left for another occupation and how many left work altogether.

Assembling it turns out to require settling a measurement question that the mobility literature has circled for two decades. Kambourov and Manovskii (2008) devoted substantial effort to spurious occupational transitions arising from independent coding, and Mellow and Sider (1983) showed much earlier that worker and employer reports of the same job frequently disagree. What has not been available is a direct measurement of how large the problem is in the survey everyone uses, on people whose occupation is known not to have changed. Section 2 provides one, and the answer is large enough to determine which instrument the rest of the paper can use.

The contribution is therefore in two parts, and the first is the condition of the second. Section 2 establishes what the Current Population Survey can and cannot record about a person's occupation, and shows that the retrospective design is the only viable instrument. Section 3 uses it to build the flow account. Section 4 reports what the account shows, and Section 5 sets out the scope of the measures.

A companion paper takes the same account to the question of whether these patterns vary between states, and the geographic movement of workers that question requires. Nothing here depends on it.

2 The instrument

2.1 Three measures of one transition

The Current Population Survey records occupation by asking what kind of work a person does, writing down the answer, and assigning a code to the description. The annual social and economic supplement, fielded each March, asks the same question twice, once about the present and once about the longest job held during the previous calendar year, and the difference between the two codes is the annual occupational transition this paper uses.

The natural alternative is to follow a person across interviews. The survey's rotation design returns a household to the sample exactly twelve months after its first interview, so the same person appears in March of one year and March of the next, with occupation collected contemporaneously on both occasions and an eight month gap between visits during which no answer is carried forward. That yields a second measure of the same annual transition, and comparing the two is informative because they agree on almost everything else. Linking the supplement to the March monthly file for the same month gives agreement on current occupation in 99.99 per cent of 504,531 people, so the two files share one coding of the present job and any disagreement across a year cannot be attributed to different instruments.

Table 1 sets the two measures against a third, which is the diagnostic. Taking last year's occupation as recorded contemporaneously a year earlier and comparing it with last year's occupation as recalled in the supplement puts two independent codings against each other on the same period, so a genuine change of occupation cannot explain a disagreement. They disagree 52.2 per cent of the time.

Table 1: Three measures of the same annual transition, 2017 to 2025

	Detailed (474 codes)	Major group (23)
Same period, two independent codings	52.2%	33.5%
Twelve months apart, two independent codings	50.0%	
Twelve months apart, coded in one interview	10.3%	3.3%

The second row is the consequence. A linked measure of annual mobility returns 50.0 per cent, which is below the disagreement between two codings of a period during which nothing could have changed, so a linked panel design applied to this survey recovers coding discordance with real movement somewhere beneath it and no way to separate the two.

2.2 The tightest version, and which instrument is wrong

A disagreement establishes that one of two measurements is wrong without saying which, and the possibility that the supplement is suppressing changes that really happened has to be taken seriously, since two codes assigned at one sitting from one description could easily be assigned the same value out of consistency rather than accuracy.

The tightest test restricts attention to people whom the supplement records as having held the same occupation in both years and who worked all fifty two weeks of the reference year. For these 114,491 people the independently coded occupation from a year earlier should be identical, and it disagrees in 48.1 per cent of cases at the detailed level, 41.1 at the level of the ninety seven minor groups and 31.8 across the twenty three major groups. Figure 1 sets that against the movement the supplement itself measures on the same people.

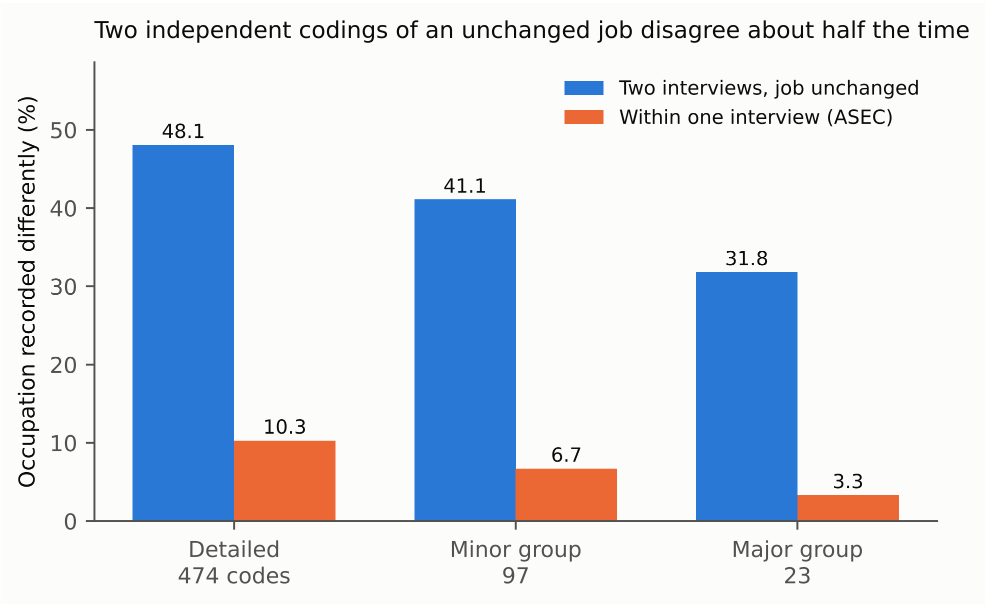

Figure 1: Two independent codings of an unchanged job disagree about half the time, while the within interview measure records annual movement an order of magnitude smaller.

Three pieces of evidence say the coding is at fault rather than the supplement. The person link is sound, because on the same people state of residence differs in 0.00 per cent of cases and sex in 0.00, while education differs in 12.6 and occupation in 48.1, which makes occupation the least reliable variable in the survey by a wide margin. The disagreement is stable at close to fifty per cent across all nine year pairs from 2016 onwards, including those either side of the 2020 change of occupation classification, so it is a structural property rather than an artefact of one revision.

The third piece is the most telling, and it concerns the shape of the disagreements rather than their size. Were the supplement quietly suppressing real changes, the disagreements would look like plausible career moves. They do not. The single largest confusion is elementary and middle school teachers against secondary school teachers, at 0.96 per cent of all discordance, followed by first line supervisors against the staff they supervise in retail and in offices, by miscellaneous managers against specific managerial categories, by nursing aides against personal care aides, and by accountants against bookkeepers. These are boundaries in the classification system where a description could reasonably be coded either way, and they are not the paths along which careers run.

The test bounds the disagreement between two independent codings of one job. It does not measure how far a single interview suppresses genuine change through the consistency of two codes assigned at one sitting, which the shape of the disagreements argues against without quantifying.

Discordance concentrates within occupational families and along the supervisory boundary, so movement between families survives it while movement within a family does not. Registered nurses against physical therapists, a pair whose apparent flow in the account looked implausible to me before this test, accounts for 0.002 per cent of discordance and is therefore not an artefact of coding. Elementary against secondary teachers, which looked implausible for the same reason, is the worst pair in the data and should be reported only at a level of aggregation that contains both.

3 The flow account

3.1 Construction

The account uses the seventeen annual supplements from 2009 to 2025, comprising 2,924,246 person records. For every occupation and year it records the number of people who held the occupation during the previous year, the number who arrived from education, from other non-employment and from another occupation, and the number who departed to another occupation or out of employment altogether. Arrivals from outside employment are identified by the previous year occupation taking the code reserved for people who did not work, and the reason they did not work distinguishes education from caring, illness, retirement and inability to find work.

The identity closes to zero people rather than approximately, because every person contributes once to each side from a single source. Nothing is calibrated to an external total and nothing is imputed.

Two construction details matter enough to state. The 2014 supplement was fielded as a split sample, with three eighths of respondents receiving redesigned income questions and five eighths the existing ones, and analysing both halves together returns weighted totals twice the size of the population. Filtering on the flag that marks the halves gives 144.8 and 145.5 million against 290.3 million unfiltered, and both halves sit on the trend between the adjacent years. Standard errors are computed from the 160 person replicate weights the Census Bureau constructs by successive difference replication, with the multiplier of 4/160 the documentation prescribes, and rates are recomputed inside each replicate rather than propagated from the errors of their numerator and denominator, since those move together.

3.2 Coverage and precision

Precision degrades quickly as occupations get smaller. Across the cells in the account the median standard error on a rate is 1.3 percentage points where at least three hundred prior year holders are sampled, 2.8 points at between one hundred and three hundred, 4.8 points between fifty and one hundred, and 10.6 points below fifty. A cell is reported here only where at least one hundred prior year holders appear, a threshold set from that relationship rather than from inspecting the estimates, which leaves 230 occupations across the full period and 172 for the years 2022 to 2025, out of the 474 the survey distinguishes. The remainder exist in the data and are too imprecise to publish.

4 What the account shows

4.1 Arrivals

Table 2 gives the account for eight occupations, averaged over 2022 to 2025 and expressed per hundred people who held the occupation the previous year. Figure 2 shows the same information with arrivals and departures on a common scale.

Table 2: Arrivals and departures per 100 prior-year holders, average of 2022 to 2025

	Arrivals from			Departures to	
Occupation	Education	Other non-work	Another occupation	Another occupation	Out of work
Registered nurses	0.7	0.7	3.4	4.4	4.9
Elementary and middle teachers	0.5	1.1	3.6	4.9	5.4
Secondary teachers	1.0	0.6	7.6	6.7	4.7
Software developers	0.7	0.5	4.1	6.7	5.5
Truck drivers	0.4	2.1	7.7	9.3	9.6
Nursing and home health aides	0.9	1.8	13.1	13.2	9.5
Cashiers	4.8	3.0	8.7	12.3	20.3

Two things in that table run against the way workforce shortage is discussed. Registered nurses have the lowest total departure rate of the large health and education occupations at 9.4 per hundred a year, while nursing and home health aides lose 22.7, and the difference between the two is concentrated in movement to other occupations rather than in departure from work. The shortage that gets discussed as a nursing shortage is, in the flows, a retention problem in the occupations adjacent to nursing.

The general pattern holds well beyond these eight. Figure 3 plots every occupation in the account, and 98 per cent lie above the line on which arrivals from other occupations equal arrivals from outside employment. Aggregating the whole matrix, arrivals from another occupation run 4.29 times arrivals from outside employment at the detailed level, 2.89 at minor group and 1.55 across the twenty three major groups, where a coder cannot confuse a nurse with a physical therapist and the surviving flow is movement between broad occupational families. Measured against arrivals from education specifically rather than from all non-employment, the ratio is 4.3 even at that coarsest level.

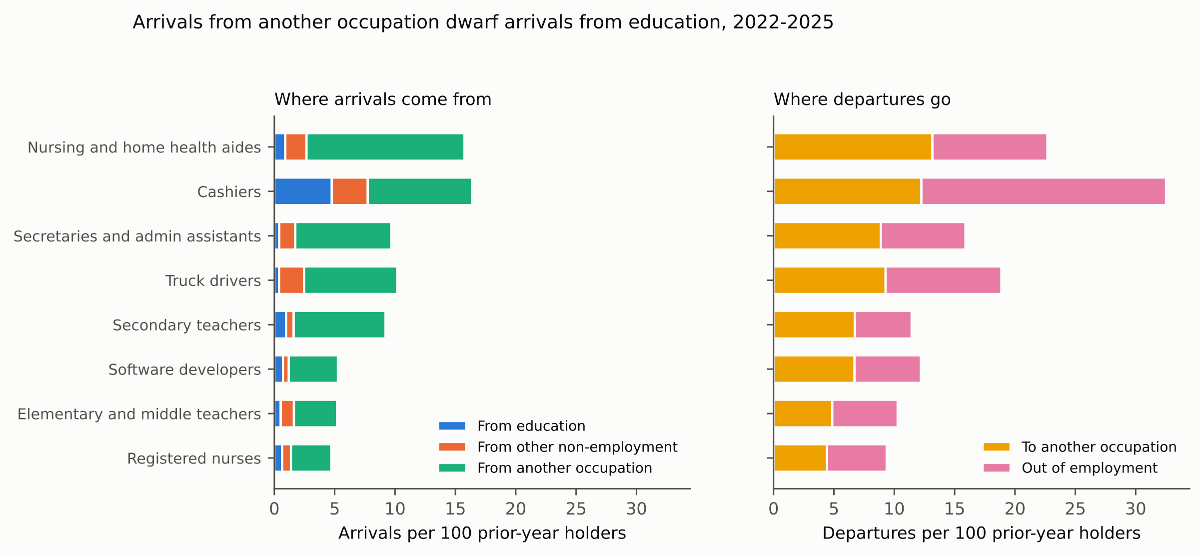

Figure 2: Arrivals from another occupation dominate arrivals from education, and the ranking of occupations by turnover runs against the way shortage is usually described.

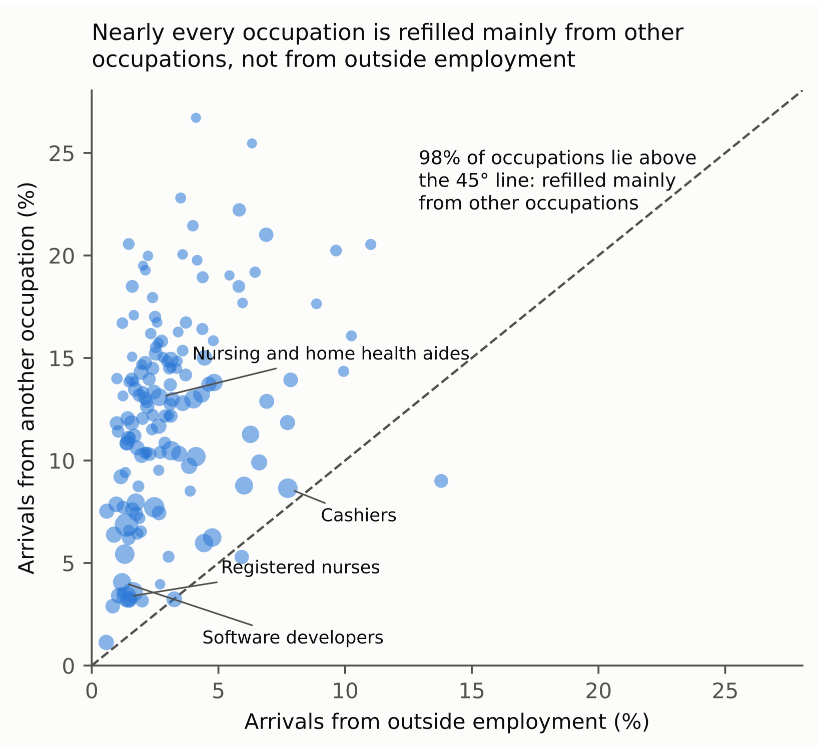

Figure 3: Arrivals from another occupation against arrivals from outside employment, one point per occupation, 2022 to 2025.

4.2 The age profile

The obvious objection is that arrivals recorded as coming from another occupation are graduates who held a job while studying, so that the training pipeline is present in the account under a different heading. If that were so the arrivals would concentrate in the youngest age bands, and they do not. Figure 4 shows arrivals from another occupation by age for three occupations, and for registered nurses the rate runs 3.1, 2.2, 3.4, 1.3 and 2.8 across the bands from twenty five upwards, which is flat rather than declining. Between 42 and 55 per cent of lateral arrivals land in the two youngest bands across the occupations examined, against roughly a third of the workforce being that age, so the skew towards youth is mild and cannot carry the objection.

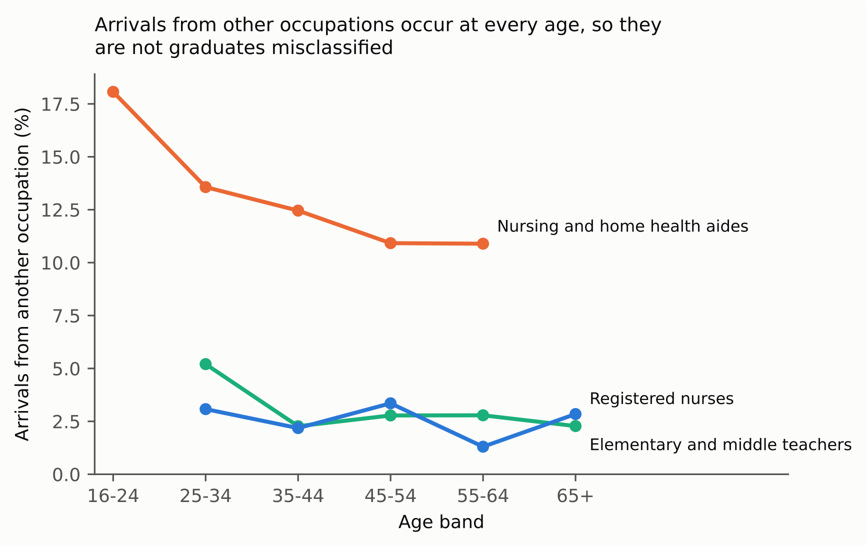

Figure 4: Arrivals from another occupation occur at every age.

4.3 Direction and churn

Gross movement between occupations is large, and most of it cancels. Splitting the transition matrix into a symmetric part and an antisymmetric one separates two way traffic between a pair of occupations from net reallocation between them, and the second cannot be produced by symmetric misclassification. At the detailed level, gross lateral movement of 18.81 million a year contains 3.43 million of net directional reallocation, a directional share of 36.4 per cent. At minor group the figures are 12.67 and 1.40 million, a share of 22.1 per cent, and at major group 6.80 and 0.49 million, a share of 14.5 per cent.

The interpretation of the symmetric remainder is bounded rather than settled, because genuine two way movement between adjacent occupations and coding discordance both live there and this decomposition cannot separate them. What it does establish is a floor on real reallocation, and the size of that floor relative to the gross flow is the point. An occupation's stock is churned by movements several times larger than the change in its size, so a policy that raises arrivals is operating against departures of comparable magnitude, and the net effect is a small difference between two large numbers.

5 Scope of the measures

Three features of the design bear on how the figures should be read. The count of people who held an occupation last year covers a twelve month window while the count of current workers is taken at a single moment, so the ratio between the two is not a growth rate, though the flows themselves and comparisons between occupations are unaffected. The survey looks back one year, so arrivals from outside employment include people returning to an occupation they had held before alongside those entering it for the first time. And a departure is the statement that a person was not in the occupation twelve months later, which a one year transition cannot separate from a temporary absence.

6 Conclusion

The account assembled here says that occupations in the United States are refilled predominantly by people arriving from other occupations, at four to twelve times the rate at which they are refilled from education, and that this holds for 98 per cent of occupations and survives aggregation to a level at which the classification cannot confuse adjacent professions. It says that these arrivals occur at every age, so they are career movement rather than training recorded under another name. And it says that most of the movement cancels, so the gross flows through an occupation are several times the net change in its size.

The measurement result that makes the account possible may prove the more durable contribution. Two independent codings of an unchanged job disagree in half of cases, which means that occupational mobility cannot be measured in this survey by following people between interviews, and that the retrospective design, whose reliance on recall makes it look like the weaker instrument, is the one that works. That is worth knowing independently of what is done with it.

What follows for policy is a question about the size of the lever rather than a recommendation. Training capacity acts on the smallest of an occupation's inflows, and the largest inflow is governed by whatever determines who can move into an occupation from another one, which is a different set of things: credential requirements, licensure, and the transferability of skills between adjacent occupations. Which of those binds, and for which occupations, is not settled here, and it is the question the account was built to make askable.

References

Cheng, S. and Park, B. (2020) 'Flows and boundaries: a network approach to studying occupational mobility in the labor market', *American Journal of Sociology*, 126(3), pp. 577–631.

Davis, S. J. and Haltiwanger, J. (2014) 'Labor market fluidity and economic performance', *NBER Working Paper* 20479.

Flood, S., King, M., Rodgers, R., Ruggles, S., Warren, J. R. and Westberry, M. (2024) *Integrated Public Use Microdata Series, Current Population Survey*. Minneapolis: IPUMS.

Kambourov, G. and Manovskii, I. (2008) ‘Rising occupational and industry mobility in the United States, 1968–97’, *International Economic Review*, 49(1), pp. 41–79.

Mellow, W. and Sider, H. (1983) ‘Accuracy of response in labor market surveys: evidence and implications’, *Journal of Labor Economics*, 1(4), pp. 331–344.

Moscarini, G. and Thomsson, K. (2007) ‘Occupational and job mobility in the US’, *Scandinavian Journal of Economics*, 109(4), pp. 807–836.